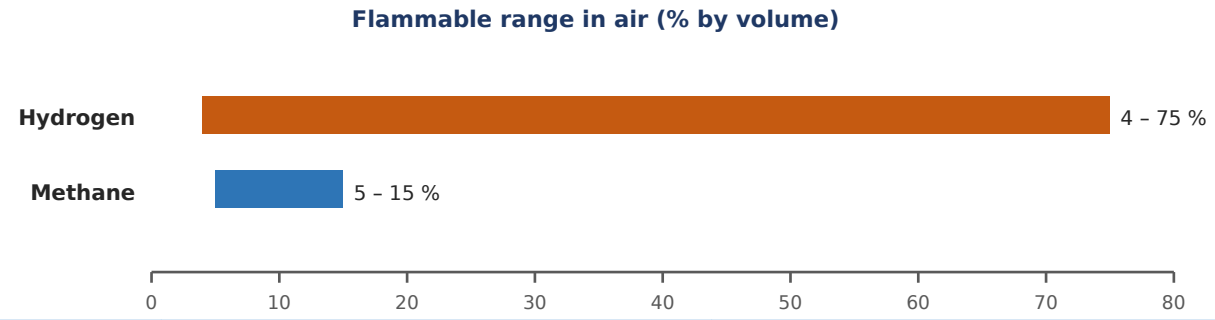
MODULE 10

Safety & Industrial Standards

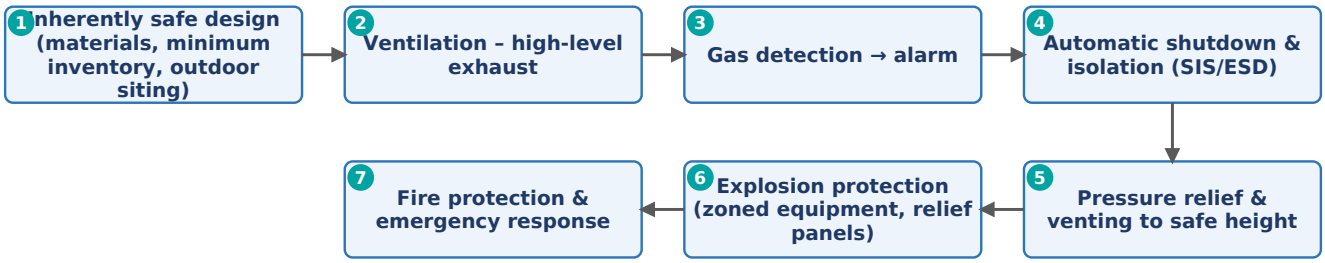
Hazards · protection layers · area classification · codes

10.1 Hazard properties



Hazard	Why it matters	Control
Flammability	Ignites from 4 % to 75 % in air; very low ignition energy (~0.017 mJ) [R6]	Leak prevention, ventilation, ignition-source control, bonding/earthing
Leakage	Smallest molecule – leaks through fittings, seals and some materials	Welded joints where possible, leak testing (helium/H_2), detectors
Invisible flame	Pale flame hard to see in daylight; low radiant heat	UV/IR flame detectors; thermal cameras
Embrittlement	Atomic H diffuses into steel and reduces ductility, especially high-strength steels and welds	Material selection (austenitic stainless, low-strength steels), hardness limits, fatigue design
Pressure	Stored energy in high-pressure vessels	Relief valves, burst discs, code-designed vessels
Cryogenic (LH_2)	Cold burns; air condensation on cold lines; boil-off pressure rise	Vacuum insulation, relief, PPE
Asphyxiation	Displaces oxygen in confined spaces	O_2 monitoring, confined-space permits

10.2 Protection layers



10.3 Hazardous area classification

Zone (IEC 60079-10-1)	Meaning	Hydrogen equipment rating
Zone 0	Explosive atmosphere present continuously or for long periods	Ex equipment for Group IIC
Zone 1	Likely in normal operation	Group IIC, temperature class T1 or better
Zone 2	Not likely; if it occurs, only briefly	Group IIC, T1 or better

Hydrogen is in gas group **IIC** (the most easily ignited group, with acetylene). Its auto-ignition temperature (≈585 °C) puts it in temperature class **T1** (equipment surface ≤ 450 °C).

10.4 Key standards and codes

Standard	Scope
ISO 22734	Hydrogen generators using water electrolysis – industrial, commercial and residential
ISO 14687	Hydrogen fuel quality – product specification
ISO 19880-1	Gaseous hydrogen – fuelling stations
ISO/TR 15916	Basic considerations for the safety of hydrogen systems
IEC 60079 series	Explosive atmospheres – area classification & Ex equipment
NFPA 2	Hydrogen Technologies Code (USA)
ASME B31.12	Hydrogen piping and pipelines
IEC 61511	Functional safety – safety instrumented systems for process industry
India	PESO approvals; Gas Cylinders Rules 2016; Static & Mobile Pressure Vessels (Unfired) Rules; MNRE Green Hydrogen Standard (2023)

Always check the latest edition and the local authority having jurisdiction.

10.5 Safety scenarios (think, then read the answer)

Scenario
A hydrogen detector in the electrolyser hall alarms at a low % of LFL during start-up.

Recommended response
Do not reset blindly. Follow the alarm response: stop ignition-source work, confirm with a portable detector, locate the leak (often new flanges/fittings), isolate and depressurise via the safe vent, purge with N_2, repair, leak-test, then restart.

Scenario
The H_2-in-O_2 analyser on an alkaline electrolyser rises at low load.

Recommended response
Classic crossover issue: at low current, gas production falls but diffusion through the diaphragm does not – the H_2 fraction in O_2 rises. Increase load or shut down per the SIS trip limit, well below the lower explosive limit.

Scenario
A contractor wants to weld a bracket on a pipe rack next to a live hydrogen line.

Recommended response
Hot work needs a permit, gas test, isolation/positive isolation of the line or a safe distance, fire watch, and continuous gas monitoring.

Scenario
A crack is found at a weld in a high-pressure hydrogen buffer vessel after many pressure cycles.

Recommended response
Likely hydrogen-assisted fatigue. Take the vessel out of service, perform NDT and fitness-for-service assessment; review material, hardness and cycle count against design.

Quiz & revision – Module 10

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Gas group for hydrogen in IEC 60079:
a) IIA **b)** IIB **c)** IIC **d)** I

Q2. Temperature class for hydrogen:
a) T1 **b)** T3 **c)** T4 **d)** T6

Q3. Where do hydrogen leaks accumulate indoors?
a) Floor pits **b)** Under roofs / high points **c)** Evenly **d)** In drains

Q4. Embrittlement is worst in:
a) Low-strength steels **b)** High-strength steels & hard welds **c)** Austenitic stainless **d)** Copper

Q5. Standard for electrolysis hydrogen generators:
a) ISO 22734 **b)** ISO 14001 **c)** IEC 61850 **d)** API 610

Answer key
Q1 – c: ; **Q2 – a**: AIT ≈ 585 °C; **Q3 – b**: buoyant; **Q4 – b**: ; **Q5 – a**: